B.E.A.C.O.N.

Bicycle Electronic Awareness and Communication On-road Network

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1.0 Motivation and Background

In recent years, bicycle transportation has experienced significant growth. This development follows the culmination of several compounding shifts in social perception, including environmental awareness, urban congestion, rising fuel costs, and an increase in personal fitness following the recovery of the COVID-19 pandemic. As the popularity of cycling has increased, the industry surrounding it has responded similarly. Changes in new legislation, city planning, and notably the popularity of E-bikes have helped to make biking more accessible, increasing the number of riders on the street. Despite these trends, improvements in biker safety have yet to match, resulting in a rise in related accidents. As more cyclists start to share roadways with motor vehicles, rider safety continues to become an increasingly important engineering concern.

Of these concerns, one of the largest contributors to bicycle-related accidents is reduced rider visibility and limited communication, which is compounded during nighttime riding or in dense traffic conditions. This is because, unlike traditional roadway vehicles, bicycles lack integrated signaling systems to communicate intentions such as braking, turning, or lane changes. Many cyclists rely solely on reflectors or hand gestures, the latter of which requires releasing the handlebars, reducing vehicle control. As a result, there is a growing demand for compact, low-power, intelligent safety systems that improve rider visibility while remaining lightweight and affordable.

Within the industry, many modern E-bikes have adopted some necessary technology to address these problems, but primarily focus on marketable features instead of safety. Consequently, we strive to meet these demands in a comprehensive manner by developing an affordable and compact bicycle system that accommodates traditional bikes, providing signaling and other support for the rider. Our interdisciplinary experience in embedded firmware development, analog and digital electronics, PCB design, power systems, sensor integration, and mechanical packaging aims to closely reflect real-world development workflows used in consumer electronics industries.

2.0 Goals and Objectives

2.1 Basic Goals

- Develop a bicycle-mounted safety system to improve rider visibility, safety, and communication with nearby drivers and pedestrians.
- Create an automatic brake light system using IMU-based deceleration sensing.
- Implement left and right turn signals using handlebar controls.
- Improve rider awareness by detecting approaching vehicles and providing warning alerts

- Design a compact, battery-powered system that mounts underneath a bicycle seat.
- Integrate sensing, processing, power management, and LED control into a single working prototype.

2.2 Basic Objectives

- Configure the microcontroller for system control and state management.
- Interface the IMU using I2C communication for deceleration detection.
- Activate brake light LEDs when measured deceleration exceeds a threshold.
- Implement left and right turn signals using handlebar switches and MOSFET-driven LEDs.
- Design regulated 3.3V and 5V supplies powered by a rechargeable battery.
- Integrate a radar sensor capable of detecting approaching vehicles and triggering an audio alarm when a vehicle is within a predefined range.

2.3 Advanced Goals

- Improve braking detection accuracy by filtering out road vibration and unwanted sensor noise.
- Increase turn signal responsiveness using interrupt-driven inputs.
- Design efficient LED driver circuitry for high-brightness daytime visibility.
- Develop a rechargeable and power-efficient embedded system with custom PCB integration.
- Reduce false brake light activation caused by bumps or sudden movement.
- Increase the accuracy and usefulness of vehicle detection by reducing false alerts and improving hazard awareness.
- Improve system reliability and durability during outdoor riding conditions.

2.4 Advanced Objectives

- Implement hardware debounce filtering and interrupt-driven input handling for reliable turn signal activation.
- Apply filtering techniques to reduce IMU noise and prevent false brake activation.
- Design high-current LED driver circuitry with PWM-based brightness and flashing control.
- Design and fabricate a compact custom PCB for sensing, control, and power circuitry
- Integrate lithium-ion charging circuitry into the power subsystem.
- Implement filtering and threshold-based processing of radar data to distinguish legitimate approaching vehicles from stationary objects and environmental noise.

2.5 Stretch Goals

- Implement auto-canceling turn signals

- Improve visibility through additional adaptive lighting features, such as automatically adjusted LED brightness
- Improve rider awareness by adding a handlebar-mounted display for vehicle detection information

2.6 Stretch Objectives

- Integrate a light sensor and dynamically adjust LED brightness
- Automatically deactivate turn signals after a specified time or after detecting a completed turn using IMU data
- Integrate a small LCD or OLED display to show radar-based vehicle distance, warning status and/or turn signal status

3.0 Existing Technologies and Related Projects

Listed in the chart are some existing technologies that aim to improve rider visibility and safety on bicycles. Using the idea behind these existing products, we aim to combine the features into one system that ensures rider safety.

3.1 Table of existing technologies

Product	Features	Cons/Comparison
Garmin RTL515	Radar Detection and Rear Lighting	No turn signal
Unit 1 Smart Light and Remote and helmet	Turn signal, rear and front light and hud overlay	Requires an expensive helmet
CYCL WingLights for bicycle	Turn signal	No rear light or brake light

3.1.1 Garmin RTL515

The first product we have listed is the Garmin RTL515, a bicycle safety device that combines a rear light with radar detection to alert riders of vehicles approaching from behind. The system improves rider awareness and visibility, but it does not include built-in turn signals(Garmin and subsidiaries *Varia™ RTL515/516*). Our project differs by focusing on both rider visibility and rider communication through integrated brake lights and turn signals.

3.1.2 Unit 1 Smart Light and Remote + Helmet

The second product we have listed is the UNIT 1 Smart Light system, which includes front and rear lighting, turn signals, and a heads-up display integrated into a smart helmet. While the system offers several advanced safety features, it requires the rider to purchase and use a specialized helmet(“Best Bike”), which increases the overall cost. In comparison, our project is designed to be a more affordable add-on system that can be mounted onto a traditional bicycle.

3.1.3 CYCLE WingLights

The third product we have listed is the CYCL WingLights system, which uses handlebar-mounted turn signal lights to help cyclists communicate turns and lane changes more clearly (“WingLights”). Although the system improves rider signaling, it does not include a rear brake light or automatic braking detection. Our design expands on this idea by combining turn signals, rear lighting, and automatic brake light activation into a single integrated safety system.

3.2 Battery Options

Battery Type	Energy Density	Weight	Safety	Cycle Life	Cost
Lithium-Ion	High	Low	Medium	500 - 1000	Low
Lithium Polymer	Very High	Very Low	Medium Low	300 - 800	Medium
Lithium Iron Phosphate	Medium	Medium	Very High	2000 - 5000	High
Nickel-Metal Hydride	Low	High	Medium High	500 - 1000	Low

4.0 Hardware/Software Design Specifications

4.1 Overall system Specifications

The table below represents the system specifications for the B.E.A.C.O.N. project. Additionally, the highlighted rows represent the three specifications intended for demonstration purposes. These traits have been agreed upon as the best features to represent the B.E.A.C.O.N. project and its capabilities.

Specification	Target Value	Verification Method

Break Detection Response Time	150 - 200 ms	Inertia Measurement Unit Timing
Turn Signal activation Latency	50 - 100 ms	Interrupt Timing Test
Break Detection Accuracy	At least 93%	Controlled Breaking Trials
Visible Signaling Distance	At least 300 ft	Outdoor Field Testing
Battery Runtime	At least 8 hours	Battery Discharge Testing
Wireless Communication Range	At least 10 ft	Range Testing
System Operating Voltage	3.3 - 5V	Multimeter Test
Weather Resistance	At least Rain Resistant	Water test
Maximum weight	At most 1 lb	Scale Measurement
Radar Detection Accuracy	At least 90%	Controlled Radar Testing

4.2 Component-Level Specifications

Component	**Specification**	**Target**
IMU Sensor	Accelerometer Range	± 16 g
IMU Sampling Rate	Data Acquisition Frequency	100 Hz
LEDs	Drive Current	350 mA
Microcontroller	Clock Frequency	80 MHz
Battery	Capacity	2500 mAh
Charging IC	Charge Current	500 mA
Wireless Module	BLE Version	5.0
MOSFET Driver	Power Efficiency	90%

Efficiency		
PCB Dimensions	Maximum Size	100 mm x 60 mm
Radar	Radar Range	at least 100 ft

4.3 System Software Diagram

The software flowchart below outlines the system process for operation. Including stop and turning signal event triggering, telemetry data for the display, and sensory alarm activation. Following the overall process the onboard microcontrollers follow to ensure rider safety and hardware reliability.

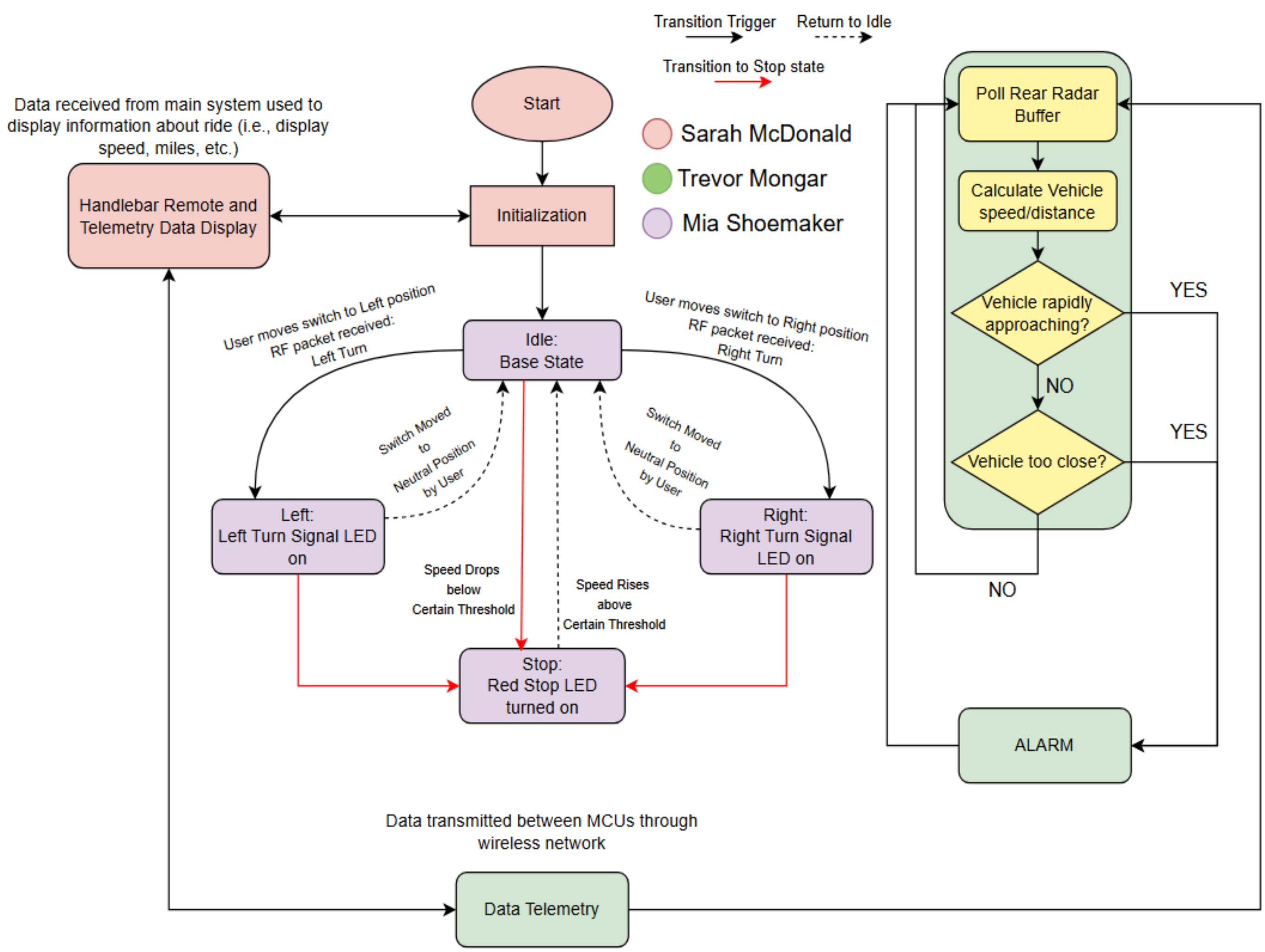


Figure 4.3 System Software Design by Authors

4.4 System Hardware Diagram

Pictured below is the overall hardware diagram for the B.E.A.C.O.N system. The diagram illustrates the core hardware components required for the B.E.A.C.O.N system to perform its expected tasks effectively and efficiently. In total, the project expects to contain two PCBs. One underneath the seat, and the other attached to the right handlebar.

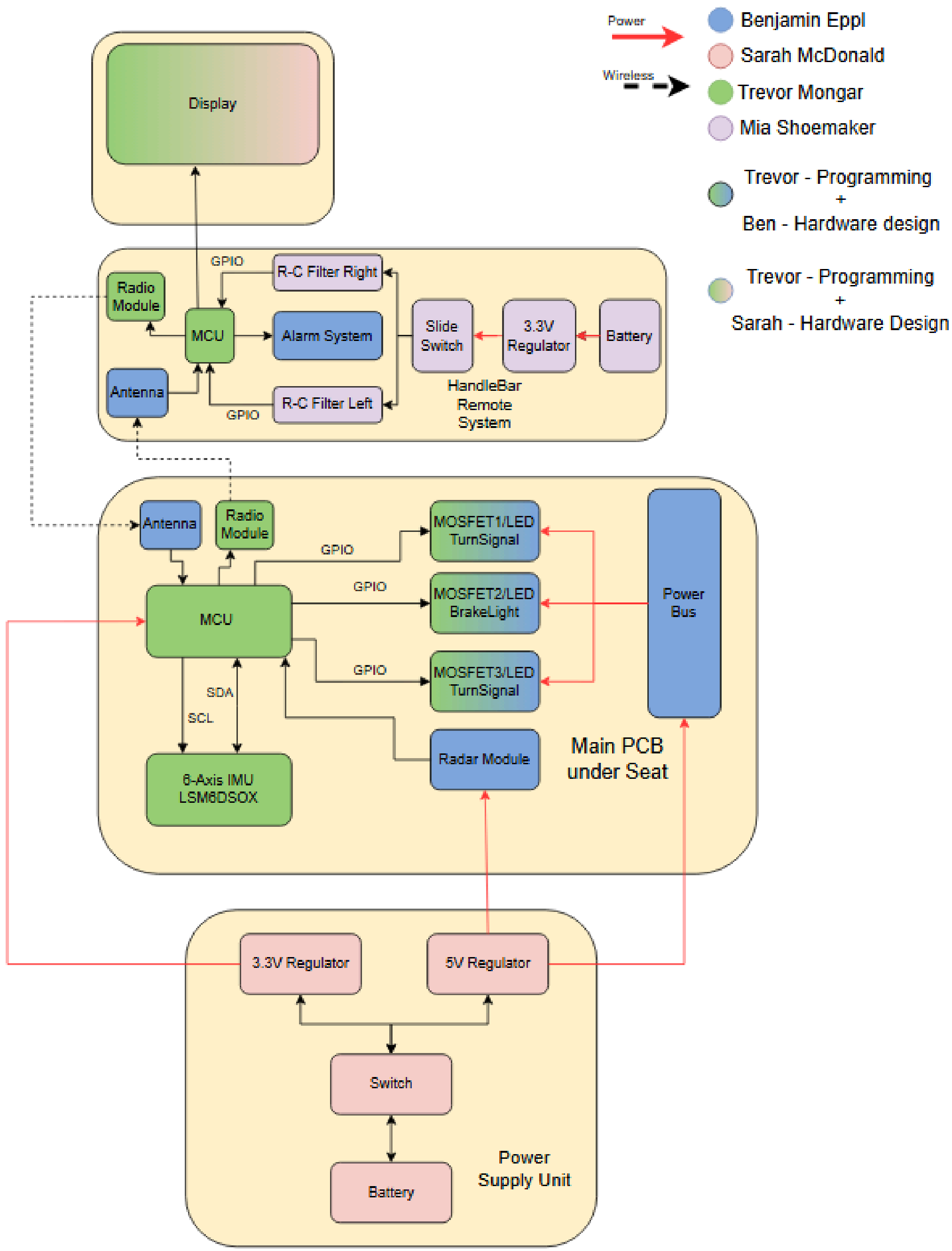
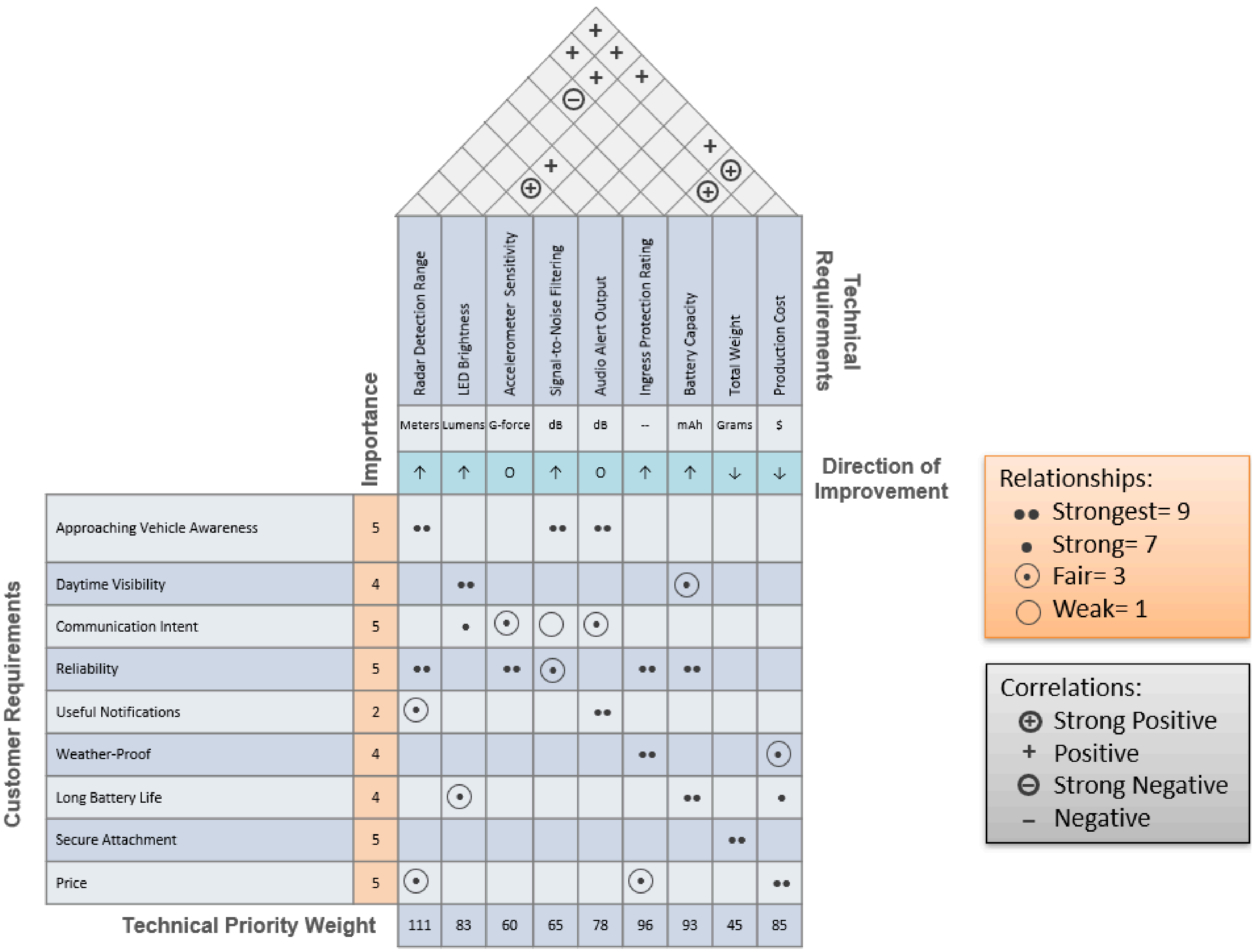


Figure 4.4 Hardware Diagram by Authors

4.5 House of Quality Diagram

Pictured below is the house of quality diagram used for this project. This diagram provides a visual aid of the engineering characteristic tradeoffs. The far left column shows common customer requirements (WHATs) versus the second row which represents the technical requirements (HOWs). By referencing this diagram, future decisions can be made towards the project on what features could be traded off or which are crucial features.



Customer Requirements	Importance	Radar Detection Range	LED Brightness	Accelerometer Sensitivity	Signal-to-Noise Filtering	Audio Alert Output	Ingress Protection Rating	Battery Capacity	Total Weight	Production Cost
		Meters	Lumens	G-force	dB	dB	--	mAh	Grams	$
Direction of Improvement		↑	↑	O	↑	O	↑	↑	↓	↓
Approaching Vehicle Awareness	5	●●			●●	●●				
Daytime Visibility	4		●●					⊙		
Communication Intent	5		●	⊙	○	⊙				
Reliability	5	●●		●●	⊙		●●	●●		
Useful Notifications	2	⊙				●●				
Weather-Proof	4						●●			⊙
Long Battery Life	4		⊙					●●		●
Secure Attachment	5								●●	
Price	5	⊙					⊙			●●
Technical Priority Weight		111	83	60	65	78	96	93	45	85

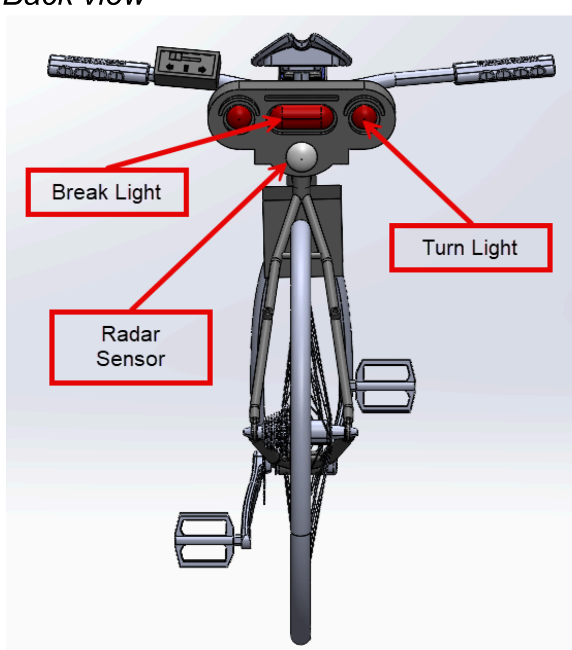
Figure 4.5 House of Quality Diagram by Authors

5.0 Prototype Renders

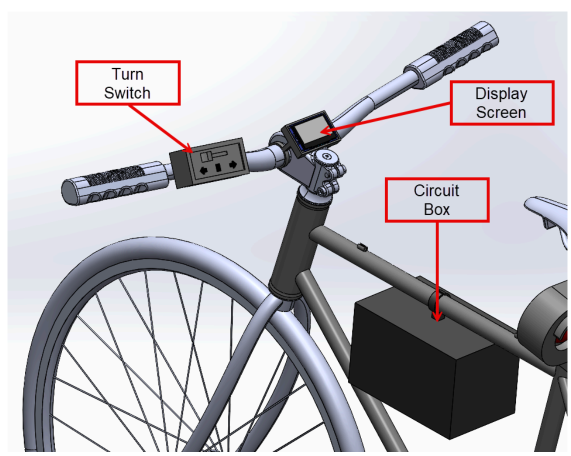
Prototype based on final design concept and open source models.

Full view

Back view



Front view



6.0 Financing and Budgeting

6.1 Bill of Materials

Component	Price	Quantity
Microcontroller1	$19.79	1
Microcontroller2	$1.17	1
IMU Sensor	$3.22	1
Distance Sensor	$11.99	1
Wireless Communication	$3.80	1
Display	$8.99	1
Brake Light	$11.99	1
Turn Light	$12.99	1
Turn Signal	$8.56	1
LED Driver	$7.99	1
PCB	Estimated $10-$20 per PCB	2
Misc.	$20-$30	Unknown
Total	$120-$140	

7.0 Senior Design Milestones

7.1 Senior Design 1 Milestones Table

Task	Description	Complete by / Due date
Divide & Conquer Report & Review Committee	Complete 10-page project proposal report along with recruiting review committee	05/29/2026

	members from ECE faculty	
Divide & Conquer Group Meeting	30-minute meeting to discuss project proposal	06/01/2026
30-Page Checkpoint	25% of Final report completed, 30 pages	06/19/2026
60-Page Checkpoint	Halfway point: Project description + Research & Investigation completed	07/03/2026
90-Page Checkpoint	75% of Final report completed, 90 pages	07/06/2026
Final Report	Final SD1 document, at least 120 pages	07/28/2026

7.2 Senior Design 2 Milestones Table

Task	Description	Complete by/Due date
Order PCB	Finalize PCB design and order any remaining components	Date TBD, Estimated time 2 weeks
Assemble PCB	Assemble any hardware and PCB	Date TBD, Estimated time 2 weeks
Software integration	Integrate braking detection, turn signals, etc.	Date TBD, Estimated time 2 weeks
Testing and Debugging	Test subsystem functionality, address any hardware/software issues	Date TBD, Estimated time 2 weeks
Prototype/Testing	Fully assemble project prototype and perform outdoor testing	Date TBD, Estimated time 2 weeks

Final Presentation	Complete Final report and Senior design showcase	Date TBD, Estimated time 2 weeks

8.0 Appendices

Aisyah Nur Jannah, et al. “Cyclist Safety in the Digital Age: A Review of Advanced Warning Technologies.” *Accident Analysis & Prevention*, vol. 217, 2025, p. 108069, www.sciencedirect.com/science/article/pii/S0001457525001551, https://doi.org/10.1016/j.aap.2025.108069.

Geneva: World Health Organization. *Global Status Report on Road Safety 2023*. 2023.

NACTO. “Urban Bikeway Design Guide.” *National Association of City Transportation Officials*, 27 June 2012, nacto.org/publication/urban-bikeway-design-guide/.

National Highway Traffic Safety Administration. “Bicycle Safety.” *NHTSA*, 18 Dec. 2018, www.nhtsa.gov/road-safety/bicycle-safety.

“Vision Zero Network.” *Vision Zero Network*, 9 Oct. 2024, visionzeronetwork.org.

Garmin. “Bike Radar and Tail Light - Varia™ RTL515.” Garmin, www.garmin.com/en-US/p/698001/. Accessed 28 May 2026.

“Best Bike Lights with Turn Signals & Brake Lights for E-Bikes”. *UNIT 1,* www.unit1gear.com/pages/smart-light. Accessed 28 May 2026.

CYCL. “WingLights Bicycle Turn Signals.” CYCL, cycl.bike/collections/winglights. Accessed 28 May 2026.

OpenAI. ChatGPT, version GPT-5.5, OpenAI, 28 May 2026, chatgpt.com/chat.